

Data Summary

Fatal Pedestrian and Cyclist Crash Risk Factors in Oregon: A Spatial & Predictive Analysis (2015-2024)

Research Question Reminder

This project investigates fatal pedestrian and cyclist crashes in Oregon between 2015 and 2024 using data from the National Highway Traffic Safety Administration’s Fatality Analysis Reporting System (FARS). Our primary research question asks: Which environmental, roadway, and temporal factors are most strongly associated with fatal pedestrian and cyclist crashes in Oregon between 2015 and 2024? Our secondary question asks whether these fatal crashes are disproportionately concentrated among certain demographic groups or communities. The data scope is now frozen and consists of a reproducible two-stage data pipeline culminating in a single analysis-ready dataset. Raw FARS crash records are first transformed into an infrastructure dataset and then enriched with U.S. Census Bureau American Community Survey (ACS) data to create the final analysis-ready dataset used throughout this project.

Data Inventory

Dataset	<i>FARS Infrastructure Dataset</i>	<i>Raw FARS Tables (Accident, Person, PBType, Vehicle, Weather)</i>
Source	NHTSA Fatality Analysis Reporting System (FARS)	NHTSA FARS National Files
License	U.S. Govt. Public Domain	U.S. Govt. Public Domain
Size	1049 rows × 30 columns	Multiple annual tables
Coverage	Oregon, 2015-2024	Oregon subset extracted from national files, 2015-2024
Refresh Policy	Annual	Annual

Repo Location	<code>data/processed/FARSinfrastructure.csv</code>	Generated into <code>data/raw/</code> during pipeline execution
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The processed master dataset was generated on June 29, 2026 from the official FARS national downloads. The repository does not store the full national raw files because of their size (tens of gigabytes). Instead, reproducible scripts recreate the Oregon-specific extracts and final analysis dataset. The intermediate infrastructure dataset is subsequently enriched with ACS demographic data to produce the final `FARSmaster.csv` analysis dataset.

Schema & Organization

Repository Structure

```

data/
├─ raw/
├─ processed/
│   └─ FARSinfrastructure.csv
│   └─ FARSmaster.csv
src/

notebooks/

deliverables/

enrich.py
initial.py
README.md

```

The project follows the DS3 repository structure established in the proposal and separates raw and processed data assets.

Primary Keys

The FARS database uses a hierarchical key structure:

- `ST_CASE` – Unique crash identifier
- `VEH_NO` – Vehicle identifier within a crash

- **PER_NO** – Person identifier within a crash
- **GE0ID** – Census tract identifier used to join crash records with neighborhood demographic characteristics

These keys allow crash-level, vehicle-level, and person-level records to be connected across multiple tables.

Join Architecture

The analysis-ready dataset is produced through a sequence of relational joins:

- Person + PType (victim demographics and positioning)
- Accident (crash environment and location)
- Weather (aggregated multi-condition weather records)
- Vehicle (speed limit, traffic control, roadway flow characteristics)
- Spatial point-in-polygon join (Latitude/Longitude → Census tract)
- ACS demographic enrichment using **GE0ID**

This process converts the highly normalized FARS structure into a single row-per-victim analytical dataset suitable for statistical analysis and machine learning.

Ingestion Status

Data ingestion is complete and stable. The ingestion pipeline consists of two scripts:

Step 1:

- Run **initial.py**
- Produces **FARSinfrastructure.csv**

Step 2:

- Run **enrich.py**
 - Adds Census tract IDs
 - Calls the ACS API
 - Appends socioeconomic variables
- Produces **FARSmaster.csv**

The pipeline has been successfully executed and produces a reproducible two-stage engineering pipeline culminating in the final analysis-ready dataset without manual intervention beyond downloading the annual FARS files. An active Census API key is required. No active ingestion development remains.

Data Quality & Profiling

Dataset Overview

<i>Metric</i>	<i>Value</i>
Rows	1049
Columns	38
Unique Crashes (ST_CASE)	490
Duplicate Rows	0

The dataset represents individual pedestrian and cyclist fatalities rather than crash events. Multiple victims may therefore share the same crash identifier.

Following enrichment, the final dataset contains both crash-level infrastructure variables and tract-level demographic variables retrieved through the ACS API. These additional fields enable neighborhood-level equity analyses while maintaining victim anonymity through Census tract aggregation.

Victim-type Distribution

<i>Victim Type</i>	<i>Count</i>
Pedestrian	881
Bicyclist	124
Personal Conveyance / Other Pedalcyclist	44

Pedestrians account for the majority of fatal non-motorist victims in the dataset.

Geographic Coverage

<i>Classification</i>	<i>Value</i>
Urban	863
Rural	185
Not Reported	1

Most fatal pedestrian and cyclist crashes occurred in urban environments.

Age Distribution

<i>Statistic</i>	<i>Value</i>
Minimum Age	1
Maximum Age	95
Mean Age	48.9
Median Age	50

The dataset covers a broad age range, indicating that fatal pedestrian and cyclist crashes affect all age groups.

Missingness & Reporting Limitations

The dataset contains very few true null values because the pipeline replaces missing values with standardized labels such as **None** / **Unknown** and **Not Reported**. However, several variables contain substantial reporting gaps.

<i>Variable</i>	<i>Not Reported Count</i>	<i>Not Reported %</i>
Traffic Control Device	593	57%
Speed Limit	248	24%
Weather	132	13%

These values are retained because they represent information unavailable in the original source records rather than data-processing errors.

Known Data Limitations

Several important limitations remain:

- FARS contains only crashes involving at least one fatality – non-fatal pedestrian and cyclist crashes are excluded.
- Traffic exposure measures (pedestrian counts, cyclist counts, vehicle miles traveled) are unavailable.
- Crash reporting practices may vary across jurisdictions and years.

These limitations will be discussed when interpreting analytical results.

Reproducibility

The analysis-ready dataset can be rebuilt using the following workflow.

Prerequisites:

- Python 3.x
- pandas
- Active Census API key

Steps:

1. Download raw FARS annual files (2015-2024)
2. Run `initial.py`
3. Insert Census API key
4. Run `enrich.py`

Ethics & Access Controls

The project relies entirely on publicly available government data and contains no personally identifiable information (PII) or protected health information (PHI). Individual victims are represented only through anonymous identifiers such as `ST_CASE` and `PER_NO`.

The primary ethical concerns are:

- Equity concerns – Historical differences in infrastructure investment and reporting practices may affect patterns we observe.
- Misinterpretation risk – Statistical associations should not be interpreted as proof of causation.
- Census variables are aggregated at the tract level and do not identify individual households, minimizing privacy risk while supporting neighborhood-level equity analyses.

Because the data is public and anonymized, no special access controls are required.

Freeze Statement

The project's data scope is now frozen. All data required to answer the primary and secondary research questions has been collected, processed, documented, and integrated into the final analysis-ready dataset `FARSmaster.csv` through a reproducible two-stage pipeline. The intermediate `FARSinfrastructure.csv` dataset is retained as an engineering artifact but is not used directly for analysis. No additional ingestion work is planned before project completion. Future work will focus exclusively on exploratory analysis, statistical modeling, spatial visualization, and interpretation of results.

The only notable change from the original charter is a shift from a Salem-focused scope to a statewide Oregon analysis. This change was formalized in the approved project proposal and reflects the availability of a complete statewide FARS dataset.